

Combining and Forward-Carrying Overlapping Wellbore Surveys: An Open Implementation Demonstrated on Public Volve Data

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2026

Version 1.1 — **2026-08-25** · concept DOI: 10.5281/zenodo.22080773 (v1.1 adds §2 trajectory reconstruction)

Abstract

When two independent surveys cover an overlapping interval of the same wellbore — most commonly a measurement-while-drilling (MWD) survey and a gyroscopic survey — the industry routinely keeps only one and discards the other. The statistical value of combining them has been recognised for over two decades (Ekseth, 1998; Bang, 2019) yet remains under-adopted, and open, reproducible implementations are scarce. This paper does two things with an open-source implementation (welleng), demonstrated end-to-end on the public Volve dataset. First, it *combines* overlapping surveys with the best linear unbiased estimator (BLUE), evaluated on the analytical arc-faithful interior covariance so the result is continuous in measured depth rather than tied to survey stations. Second, it *carries the calibration forward*: where a gyro overlaps the upper part of an MWD run, the gyro constrains the MWD’s systematic (magnetic) error — including terms a magnetic survey cannot observe from its own data — and, because that systematic realisation persists down-hole, the constraint reduces the MWD covariance in the deeper, gyro-free section. On Volve well F-4 the combination reduces the vertical 1-sigma by a factor of 1.4 over the overlap; carrying the gyro-observed declination forward reduces the deep MWD-only lateral 1-sigma unconditionally, and by roughly a factor of two when the same MWD tool is reused below. Neither method is novel — both adopt established estimation theory — so the contribution is the open implementation, the analytical/continuous formulation, and the quantified demonstration on public field data.

1. Introduction

The ISCWSA/OWSG error model (Williamson, 2000; ISCWSA Rev 5) propagates a survey tool’s error sources into a per-station position covariance. It treats a single survey; it says nothing about what to do when two surveys of the same hole overlap. In practice a definitive survey is assembled by *selecting* the best tool over each interval and concatenating (a tie-chain), discarding the redundant overlap data.

That the redundant data carries recoverable accuracy is long established. Ekseth (1998) called for “a statistical adjustment theory when redundant surveys are available”; Bang (2019, SPE-195621) sets out the rigorous weighted-averaging method for combining overlapping surveys and a combined-IPM construction with inclination/azimuth-dependent weights (building on earlier weighted-averaging work it reviews); and the ISCWSA record notes the method still “prevents wider adoption” for want of practical, trusted tooling (60th ISCWSA meeting, 2016). Separately, using an independent gyro to constrain an MWD’s systematic error via statistical estimation was presented at the 49th ISCWSA meeting (2019), where a magnetic declination error of about 0.5 degrees was recovered by comparing an MWD azimuth against a gyro.

The estimation itself is textbook — the same inverse-covariance fusion and Kalman/Gaussian conditioning used routinely in multi-sensor data fusion, geodetic least-squares network adjustment, meteorological data assimilation and aided inertial navigation. What has limited its adoption in wellbore surveying is therefore not the mathematics but three practical barriers: the **correlation bookkeeping** it requires (identifying which error sources the two surveys share, so the shared part is not wrongly reduced — the recurring “correlation is the hard part” caution in the ISCWSA record), the **select-and-concatenate convention** and a well-founded caution about *shrinking* an ellipse of uncertainty, and the **absence of open tooling**. This paper

addresses all three.

It is worth stressing that the correlation bookkeeping is not new machinery. It is the *same* shared-error accounting already performed when survey legs are **tied**: joining two sequential runs requires tracking which error sources (the geomagnetic/declination and reusable-sensor systematics) are common across the tie so the downstream leg inherits the correlated uncertainty. Standard survey software — including COMPASS, whose tie handling separates global, systematic and well-by-well terms — already carries this decomposition. What combination and forward-carry add is not a new information requirement but a *different use* of the same shared-term structure: to fuse an overlap (the cross-covariance C of Section 2) rather than to concatenate. The gap is therefore one of convention and tooling, not of theory or of missing information — a survey system able to tie legs already holds what is needed to combine them.

What is missing is not the theory but an *open, reproducible* implementation and a demonstration of the achievable benefit on public data. This paper supplies both, for two operations:

- **Combination** (Section 2): a continuous BLUE fusion of two overlapping surveys.
- **Forward-carry** (Section 3): conditioning a survey’s systematic error on an independent overlapping reference, then carrying the reduced systematic into a reference-free section — the aided-inertial-navigation “calibrate then coast” pattern applied across survey tools.

Both are demonstrated on Volve well F-4 (Section 4) and shipped in welleng (Section 5).

2. Combining overlapping surveys

Two surveys A and B provide independent position estimates over a common interval, with per-station covariances Σ_A and Σ_B . The combined best linear unbiased estimate has covariance

$$\Sigma_c = \Sigma_A - (\Sigma_A - C) (\Sigma_A + \Sigma_B - C - C^T)^{-1} (\Sigma_A - C)^T, \quad (1)$$

where $C = \text{cov}(x_A, x_B)$ is the cross-covariance from the error sources the two surveys *share*. Equation (1) is the standard best linear unbiased estimator for two correlated estimates of one quantity (the cross-covariance / “tracklet” fusion form, Bar-Shalom et al., 2001, §8.4). Error components that are fully common-mode do not reduce; only the independent part is averaged down. When the surveys are independent ($C = 0$ — the case flagged for every Volve survey tool, whose **correlate** attribute is uniformly “N”), it collapses to the classic inverse-variance information add $\Sigma_c = (\Sigma_A^{-1} + \Sigma_B^{-1})^{-1}$ — the weighted-averaging method of Bang (2019). It is BLUE, not a Kalman filter (there is no time or sequence — just two estimates of one position).

The two surveys need not be different tool *types*. Two MWD runs combine the same way — for example when a bottom-hole assembly is pulled part-way through a section and a new one re-surveys the already-drilled interval (typically while tripping back to bottom). Because both are magnetic, their **geomagnetic reference error is shared** (same location, time and model) and is kept common-mode through C — it does not reduce; only the **independent** parts (the two tools’ sensor biases and the two BHAs’ interference and sag) average down. The gain is therefore real but smaller than an MWD+gyro combination, where the gyro — being declination-independent — also collapses the geomagnetic term. The same C machinery expresses both cases. *When* the re-survey is run matters, and carries its own value — see Section 3.

Two surveys rarely share exactly the same station depths, so both must be brought to a common measured depth. Linearly interpolating the assembled covariance between stations under-reports the uncertainty — and hence the separation factor — near doglegs. welleng instead evaluates each survey’s covariance at any measured depth **analytically**: each error source is propagated onto the minimum-curvature arc through the partial-leg Rodrigues rotation — the same Rodrigues form of the minimum-curvature interpolation used in welleng’s **MinCurve** (Sawaryn and Thorogood, 2005) — so the covariance is an analytical function of arc position rather than an interpolation of the covariance matrix. The combination is therefore continuous and dogleg-faithful — queryable at any depth in the overlap, on or off a survey station. (The magnitude of the linear-interpolation error is quantified in a companion welleng result on the analytical interior covariance.)

The implementation validates against an independent stacked generalized- least-squares derivation to a relative tolerance of 10^{-8} (covariance and position), and against a Monte-Carlo oracle (Figure 3A). As an external check, it reproduces the published result of Bang (2019, SPE-195621), whose “rigorous averaging” is precisely this $C = 0$ fusion: combining two independent surveys of similar accuracy reduces the one-sigma by the equal-accuracy limit $1/\sqrt{2}$ — a 29.3% reduction, matching Bang’s stated “up to 29%”. The larger lateral reductions Bang reports (his combined lateral extent about a third of the magnetic-only value) arise

from a *realistic vendor* continuous-gyro model whose lateral accuracy exceeds the MWD’s; the standard IS-CWSA/OWSG gyro model is by design conservative — poorer than the MWD model — so the gain realised in practice depends on the actual gyro’s quality, and the demonstration below, which uses the survey tools’ own exported error models, is correspondingly a conservative floor.

Reconstructing the fused trajectory as a survey

The combination returns a covariance and a best-estimate position at each measured depth, in the north-east-vertical frame. Conventional survey software and reporting, however, expect a minimum-curvature (measured depth, inclination, azimuth) listing, and recovering one from the fused position path was raised as a practical obstacle. It is in fact analytical: with the tie-in tangent fixed, each leg’s end tangent is the start tangent *reflected about the chord*,

$$\hat{t}_{i+1} = 2(\hat{c} \cdot \hat{t}_i)\hat{c} - \hat{t}_i,$$

which is the minimum-curvature chord-bisector property. The tangents therefore march out in a single pass with no iteration (welleng’s `survey_from_positions`).

Two observations frame the reconstruction. First, the fused stations are **calculated, not measured**: the fused position is a derived best estimate, so the whole listing is a calculated product — there is no measured station that must be preserved verbatim. Measured depth, by contrast, *is* measured, and is held exact: it labels the reconstruction, and the single-pass march above returns a minimum-curvature listing at the query depths whose dogleg severity tracks the input surveys. Second, the fused positions do not, in general, lie on a single minimum-curvature arc chain at the query depths: averaging two surveys’ positions leaves some consecutive fused points marginally farther apart than the query interval permits and others marginally closer (in the F-4-style case, roughly a 45/55 split, each by up to a few centimetres). With measured depth held exact, the reconstruction therefore carries a small position residual against the raw fused points — centimetre-scale per station, accumulating to a decimetre-scale closure at the segment end. This is a genuine consequence of fusing positions, not an error, and it sits three orders of magnitude below the metre-scale position uncertainty the survey carries in the first place.

Where the fused overlap is interior — its last station is not the well’s total depth, the common case — the reconstructed segment must join a single-source continuation beyond the overlap. That continuation is a real measured survey (the longer of the two), so the seam is pinned to it: the reconstruction’s final target is the continuing survey’s position interpolated at the seam measured depth, and the accumulated closure is absorbed in the final section — extending or truncating the last leg’s chord to land on that point — so the fused segment and the continuation form one continuous minimum-curvature path with no position jump. This costs the (sub-uncertainty) fusion gain at the single seam station; the reconciliation direction is data-dependent (both signs occur across wells) and its magnitude is well below the position uncertainty. Because the fused covariance is a continuous function of measured depth (Section 2), it is evaluated at *every* station, so each carries its own true ellipse of uncertainty. The result is a self-consistent fused survey — (measured depth, inclination, azimuth, covariance) at every station — drawable as a minimum-curvature well path and carrying a true covariance throughout.

The single-arc form is exact when each leg is representable as one minimum-curvature arc; its residual grows with the curvature *change* across a leg (build-to-turn transitions, dogleg gradients), so it is a function of trajectory shape rather than a clean law in dogleg severity and station spacing. For a typical build (2–3°/30 m) it stays a small fraction of the EOU at ordinary station spacing — of order 1% at ~30 m, within a couple of per cent out to ~120 m — and becomes a large fraction (metres in absolute terms) only beyond ~300 m station spacing, a regime in which the survey is under-sampled regardless. Where wider spacing or a sharp transition is encountered, the remedy is to sample the *continuous* fusion (Section 2) on a finer measured-depth grid so each sub-leg is again a single arc, at the cost of calculated inner stations.

This reconstruction is presented as a **concept pending field validation**: it is verified here on synthetic and Volve data and is analytically well-posed, but has not yet been exercised across a range of real wells with their tie-on and continuation practices, which is where its edge behaviour (the closure reconciliation above, near-tangential legs, coarse or transition-heavy sections) should be confirmed.

3. Carrying the calibration forward

A magnetic MWD survey cannot observe some of its own systematic errors from its own data — magnetic declination and certain sensor biases are degenerate in a magnetic-only least-squares solution. An independent

gyro, being declination-independent (it references Earth’s rotation and gravity, not the geomagnetic field), breaks that degeneracy: over an overlap it observes the MWD’s realised systematic error directly.

Crucially, that systematic realisation is *the same* over the whole tool run — it persists into the deeper section the gyro never reached. So the overlap observation reduces the MWD covariance at depth. With the target’s shared systematic sources giving the cross-covariance $\text{cov}(x_{\text{deep}}, z)$ between a deep station and the overlap observation $z = x_{\text{MWD}} - x_{\text{gyro}}$, the conditioned deep covariance is the Schur complement

$$\Sigma'_{\text{deep}} = \Sigma_{\text{deep}} - \text{cov}(x_{\text{deep}}, z) \text{cov}(z)^{-1} \text{cov}(z, x_{\text{deep}}), \quad (2)$$

with $\text{cov}(z) = HH^T + \Sigma_{\text{gyro}}^{\text{overlap}} + \Sigma_{\text{MWD}, \text{random}}^{\text{overlap}}$ and H the target systematic sources’ propagation over the overlap. Equation (2) is the Gaussian conditional (Schur-complement) covariance — a single Kalman update step (Bar-Shalom et al., 2001, §5) — i.e. error-state / aided-inertial- navigation conditioning of a persistent bias state; the joint-covariance Schur form is used because it inverts the well-conditioned $\text{cov}(z)$ rather than the ill-conditioned observation-noise matrix. The concept is that of the 49th ISCWSA meeting (2019); the method is standard aided navigation (GPS-denied “coasting” of a calibrated inertial system).

Only sources that persist carry, and persistence is per class. A source is correlated between the overlap and the deep section — and therefore carries — only if it is the *same realisation* in both:

- **Geomagnetic reference (declination, field strength, dip)** is a property of the field model at that location and time, not of the tool or the bottom-hole assembly (BHA). It persists across *any* bit trip, BHA change, or tool change, and it is precisely the error an independent gyro observes best. This carry is therefore **unconditional**.
- **Sensor bias, scale and misalignment** are tool-borne: they persist only if the **same MWD tool** (sensor package) is reused deeper. A new BHA is fine so long as the tool is the same; a *different* tool re-realises these and they do not carry.
- **BHA-borne terms (axial magnetic interference, sag)** re-realise with a new BHA and do not carry across one.

The implementation selects which classes persist (`persist='global'` for the unconditional declination carry; `persist=('global', 'systematic')` when the tool is reused); a non-persisting source has independent deep/overlap realisations, contributes nothing to the cross-covariance, and is left at its full value.

Observability is also geometry-dependent. The overlap constrains only the modes it *excites*; a term that matters at depth but is not exercised by the overlap geometry (a different inclination/azimuth regime) is not carried. The benefit is bounded by the overlap’s observability, not its length alone.

Timing the re-survey — retrospective versus prospective. The forward-carry recasts an operational choice. A re-survey run while tripping *out*, after the section is drilled, improves only the *final* definitive — a retrospective gain. A re-survey run while tripping *in* — the overlapping pass an independent tool (gyro, or simply a fresh MWD) makes over the already-open hole *before* drilling ahead — lets the reduced systematic be **banked and carried forward into the interval about to be drilled**. The tighter uncertainty is then available in real time, while it can still change decisions (anti-collision separation, target sizing, an intersect approach), not merely at the end of the well. Where survey accuracy is a live drilling requirement rather than an end-of-well record, the way-in overlap is worth its rig time precisely because Equation (2) makes the gain prospective. This is the same combine-then-carry machinery; only *when* the overlap is acquired changes what it can influence.

The implementation is validated by an independent Monte-Carlo gate in which the systematic **re-realises** between overlap and deep (a different tool), so only the persisting (global) source may carry: simulating the true target and reference errors, forming the overlap observation, and estimating the deep error with the BLUE gain, the empirical residual covariance matches Equation (2) (Figure 3B).

4. Demonstration on public Volve data (well F-4)

Volve well F-4 carries an MWD survey over the whole run and a stationary gyro (“Wellbore Surveyor”) overlapping the upper part, with MWD continuing alone below about 2640 m. All error models are the exact per-tool IPMs imported from the public Volve EDM export; the geomagnetic reference is read from the export’s DP_MAGNETIC record, so magnetic inputs are identical to the operator’s. The whole demonstration is reproduced by `examples/survey_combination_volve_f4.py` (Section 5).

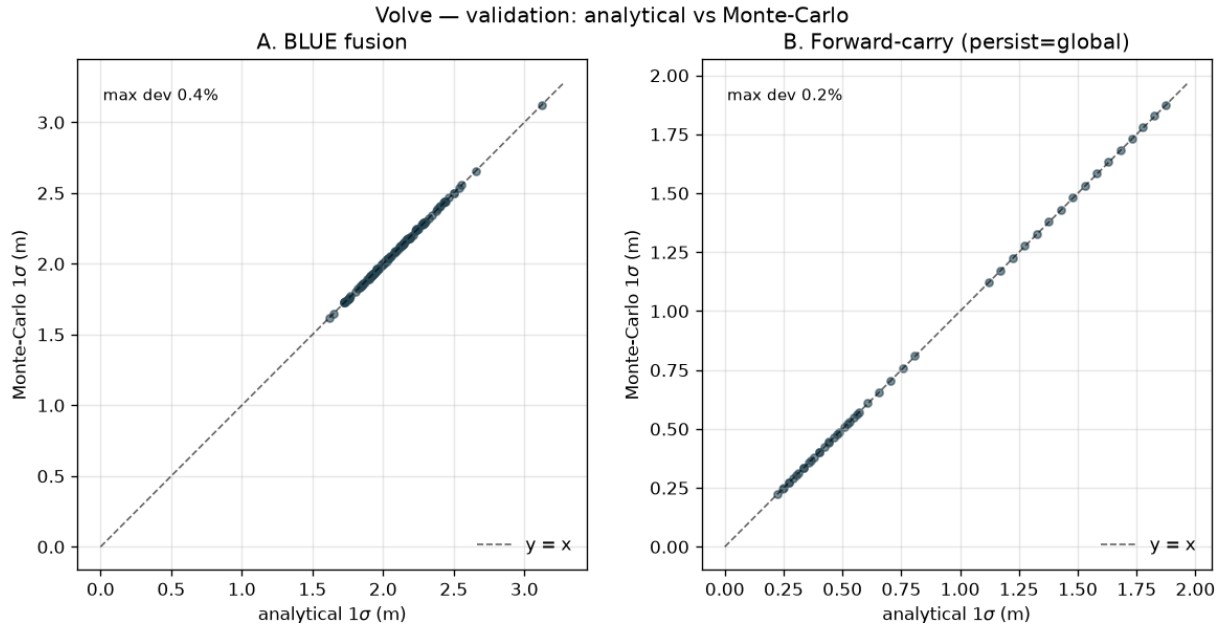


Figure 1: Figure 3. Validation — analytical prediction versus Monte-Carlo empirical 1-sigma, on the line $y = x$. (A) BLUE combination over many random shared/independent covariance cases. (B) forward-carry with the systematic re-realising between overlap and deep (persist = global). Both lie on $y = x$ within Monte-Carlo noise.

Combination (Figure 1). Over the overlap the fused 1-sigma sits below both input surveys on every axis. At the base of the overlap the reduction relative to the better single survey is 1.2 (north), 1.3 (east) and 1.4 (vertical).

Forward-carry (Figure 2). Two cases, by what is assumed to persist. The **declination carry alone** — unconditional, requiring no assumption about the deeper BHA or tool — reduces the deep lateral 1-sigma by about 1.2 (north, east) at total depth. If the **same MWD tool is reused** below (a new BHA is fine), the sensor systematic carries too and the deep lateral 1-sigma falls by roughly a factor of two — 2.0 (north), 2.2 (east). The reduction persists through the entire gyro-free section, because it acts on the systematic realisation the gyro constrained, not on any measurement the gyro made at depth. (In F-4 the deep section crosses the 12-1/4-inch to 8-1/2-inch hole-size change, so the same-tool case assumes the operator re-ran the same MWD sensor package below the shoe — common practice, but an assumption; the declination carry needs no such assumption.)

5. Open implementation

The methods are in welleng (open source):

- `welleng.combination.fuse_covariances` — BLUE fusion (Equation 1), with the cross-covariance for shared terms and a positive-semidefinite guard.
- `welleng.combination.combine_surveys` — continuous combination via the analytical interior covariance `Survey.err.cov_nev_at`; with `return_trajectory=True` it also reconstructs the fused (md, inc, azi) listing (single-arc, measured depth held exact), carrying the covariance at every station.
- `welleng.utils.survey_from_positions` — the analytical (reflection) inverse of the minimum-curvature method: (md, inc, azi) from a fused position path and a fixed tie-in tangent.
- `welleng.combination.carry_systematic_forward` — the Schur conditioning of Equation (2).

Each is a scalar reference implementation (one survey pair in, one result out), with the correctness gates described above kept as regression tests.

Every number and figure in Section 4 is reproduced by the shipped example `examples/survey_combination_volve_f4.py` (python `examples/survey_combination_volve_f4.py`, add `--plot` for the figures). It uses only public data: the F-4 survey, the two exact per-tool IPM error models and the magnetic reference are packaged with the repository (derived from the public Volve EDM export, CC BY 4.0), so no external download is required.

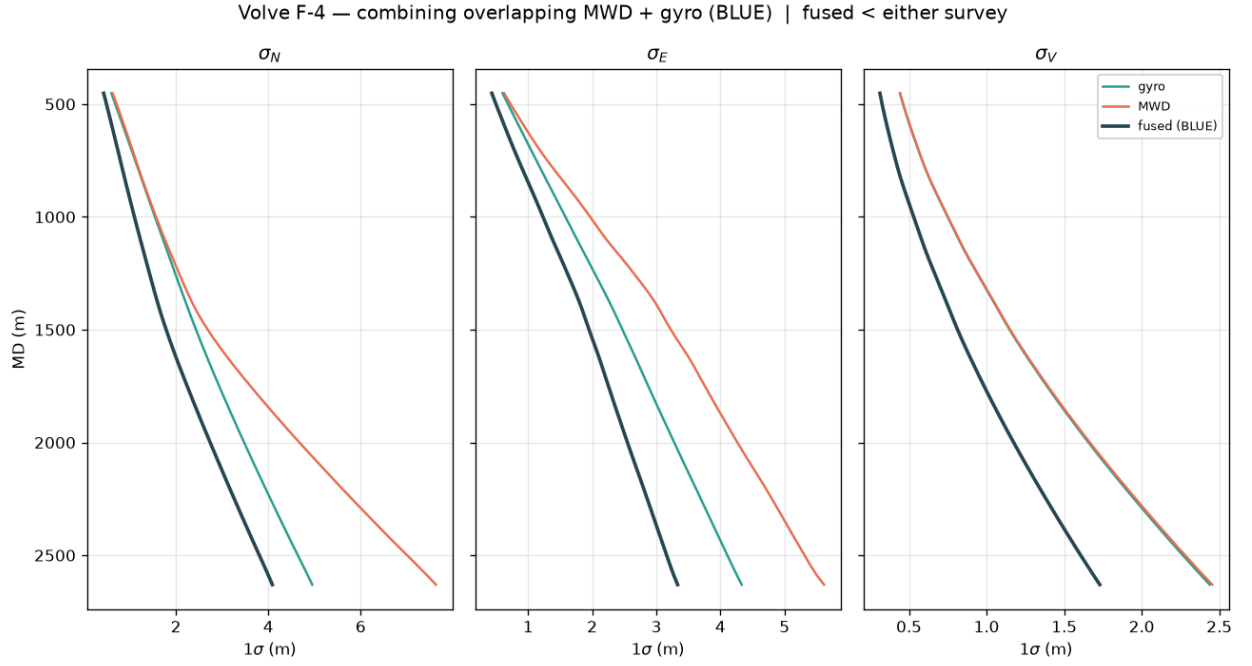


Figure 2: Figure 1. Combining overlapping MWD and gyro on Volve F-4 (BLUE). The fused curve is below either input on all three axes.

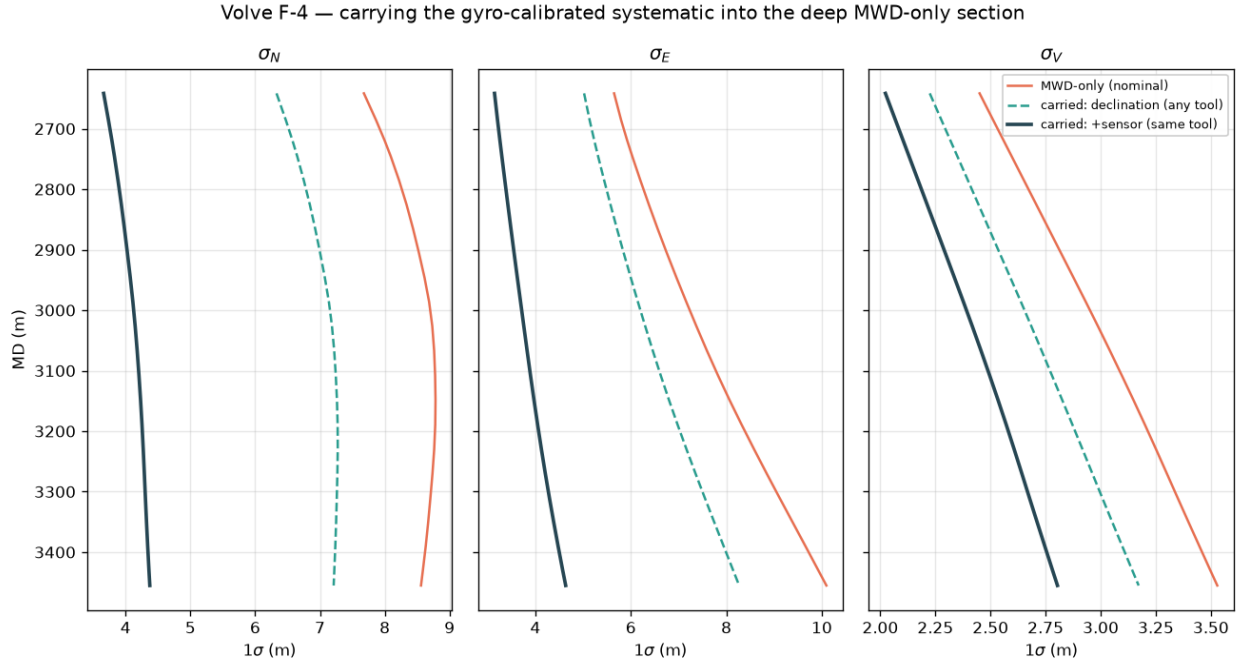


Figure 3: Figure 2. Carrying the gyro calibration into the deep MWD-only section on Volve F-4. Dashed: declination only (unconditional). Solid: plus sensor systematic (same MWD tool reused). Both stay below the nominal MWD-only covariance throughout.

6. Limitations

The combination assumes the shared/independent decomposition between the two tools is modelled correctly; the Volve demonstration uses the export’s own `correlate = N` (fully independent), which is optimistic where the two tools in fact share a tie-on or grid-convergence term. The forward-carry benefit is observability-bounded and persistence-bounded (Section 3): only the geomagnetic reference carries unconditionally, while the sensor systematic carries only if the same MWD tool is reused below — an operational assumption the analyst must confirm, and one the EDM export does not record (it carries no tool serial). Both are demonstrated on one public dataset; broader field validation is future work.

7. Suggestions and future directions

The following are **suggestions**, not validated results — offered because they follow naturally from the mechanism, but each needs work before it could be claimed.

- **Shared-surface-location declination (an IFR-from-gyro suggestion).** Wells sharing a surface location — a platform, a subsea template, or a land pad — share essentially the same geomagnetic model error, so a gyro run on the first well constrains a declination error common to the others, and the correction could in principle benefit them too. This is a form of **in-field referencing (IFR)** — established practice, usually from a magnetometer field survey — derived instead from a gyro’s true azimuth. The comparison must be precise (Jamieson, ISCWSA eBook): standard IFR’s deliverable is the **crustal correction**, which “vary only on geological timescales and therefore can be considered fixed over the lifetime of the field” — so it is **not** time-dependent, and neither is a gyro-derived equivalent over that field. The main-field secular variation is tracked separately by the geomagnetic reference model (BGGM in commercial workflows; welleng’s own client uses the free WMM/IGRF, and for this reproduction reads the operator’s stored reference from the EDM directly). The genuinely time-variant part is the **diurnal / ring-current disturbance**, corrected in real time by **IIFR** (interpolated IFR). A gyro therefore captures the total azimuth error (crustal + main-field + the diurnal *at its run time*) directly, and could substitute for the **crustal** IFR correction over a platform while also delivering an independent survey and the forward-carry — but it does **not** replace **IIFR’s real-time diurnal correction** for wells drilled at other times, nor IFR’s **spatially-mapped** crustal correction over a large field. Realising the platform-scale idea would require sharing the geomagnetic- reference source across wells’ error models; it should be framed against existing IFR/IIFR methods, not as a new concept.
- **Cost (an open question).** Where a gyro is run for its own sake, the declination information is effectively a byproduct; leveraging it across contemporaneous platform wells could be a cost-effective complement to a dedicated IFR service. Whether it is cheaper overall is operator- and case-specific (a standalone gyro run is rig time versus an IFR data service) and is not assessed here.
- **Correlation modeling.** The Volve demonstration uses the export’s `correlate = N`; a per-error-source shared/independent map between the two tools (tie-on, grid convergence) would refine the combination where it is not.

8. Conclusion

Combining overlapping surveys, and carrying an independent survey’s calibration forward into a reference-free section, recover substantial wellbore-position accuracy that current select-and-discard practice leaves on the table — a factor of about two on the deep lateral uncertainty in the Volve F-4 example. The estimation is textbook; the contribution here is an open, reproducible implementation, an analytical continuous formulation, and a quantified demonstration on public field data.

References

- Bang, J. (2019). Practical Method to Benefit from the Improved Accuracy of Combining Overlapping Wellbore Surveys. SPE-195621-MS. <https://doi.org/10.2118/195621-MS>
- Bar-Shalom, Y., Li, X.-R., Kirubarajan, T. (2001). Estimation with Applications to Tracking and Navigation. Wiley. ISBN 978-0-471-41655-5. [BLUE/cross-covariance fusion (§8.4); Kalman/Gaussian conditioning (§5)]
- Codling, J. (2018). Survey-Error Distribution and Its Effect on Collision Avoidance. IADC/SPE-189654-MS. <https://doi.org/10.2118/189654-MS>
- Ekseth, R. (1998). Uncertainties in Connection with the Determination of Wellbore Positions. PhD thesis, Norwegian University of Science and Technology (NTNU), Trondheim. ISBN 82-471-0218-8.
- ISCWSA (2016). 60th Meeting — Precise Separation Factor calculation proposal. <https://www.iscwsa.net>
- ISCWSA (2019). 49th Meeting minutes — combination of gyroscopic and magnetic survey data (statistical estimation of declination). <https://www.iscwsa.net>
- Sawaryn, S. J., Thorogood, J. L. (2005). A Compendium of Directional Calculations Based on the Minimum Curvature Method. SPE Drilling & Completion 20(1): 24–36. SPE-84246-PA. <https://doi.org/10.2118/84246-PA> [Rodrigues form of minimum-curvature interpolation]
- Torkildsen, T., Håvardstein, S. T., Weston, J. L., Ekseth, R. (2008). Prediction of Wellbore Position Accuracy When Surveyed With Gyroscopic Tools. SPE Drilling & Completion 23(1): 5–12. SPE-90408-PA. <https://doi.org/10.2118/90408-PA>
- Williamson, H. S. (2000). Accuracy Prediction for Directional Measurement While Drilling. SPE Drilling & Completion 15(4): 221–233. SPE-67616-PA. <https://doi.org/10.2118/67616-PA>

(*welleng software concept DOI 10.5281/zenodo.20968887.*)